Psychology - Structural Correspondence with the Five-Stage Model

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Psychology

Structural correspondence survey with the five-stage model (Ba, Nami, En, Uzu, Taba)

Survey Overview

- **Scope**: 10 major theories in psychology
- **Research question**: Do theories in psychology structurally correspond to the five-stage model?
- **Results**: Conditional correspondence 1

Overview of the Five-Stage Model

Stage	Definition
Ba (Field)	An undifferentiated state. Initial conditions where neither direction nor structure has yet been determined
Nami (Wave)	A stage of exploration where multiple directions diverge and compete
En (Edge)	A state of tension where opposing elements coexist without converging to either side. A place where elements meet at boundaries, influence each other, and relationships emerge
Uzu (Vortex)	A stage where new coherence (order) spontaneously arises from within tension
Taba (Bundle)	A stage where form is established and stabilizes as a reusable structure

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Overall Structural Correspondence

Figure 1: Domain — Research Overview Infographic

Temperature	Theory group	Assessment
Near-definitive	Insight problem solving, Bayesian brain hypothesis, schema theory, bias competition in attention, cognitive flexibility	High correspondence across all five stages, with concrete descriptions of the structure of En
Probable	Cognitive dissonance, flow experience, conceptual metaphor	Correspondence exists, but some compression or automaticity remains at certain stages
Conditional	4E cognition, affordance	As metatheoretical frameworks, they describe the nature of Ba rather than specific processes

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Key Entry 1: Insight Problem Solving (Ohlsson; Knoblich)

- Insight problem solving is a theory that describes the transition from impasse (being stuck) to sudden solution. Ohlsson (1992) proposed a model in which problem solvers fall into an impasse because they are constrained by their initial representation (the initial way of grasping the problem), and a solution emerges when the representation changes through constraint relaxation or re-encoding. Knoblich et al. (1999, 2001) demonstrated with matchstick problems that task difficulty depends on "the depth of constraints to be relaxed" and "the strength of chunks to be decomposed."
- **As fact**: Knoblich et al. (2001, N=24) confirmed, through eye movement tracking, that the moment when the solver's attention shifts from "value" to "operator" corresponds to representational change. Additionally, Hattori et al. (2013, N=509) showed that subliminal hints (hints at a level below conscious awareness) improve solution rates, demonstrating the existence of constraint relaxation without awareness.
- **As reading**: Here we read the process of qualitative transformation of cognitive structure in which, after being stuck through initial representation, a new representation is established through release

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

of constraints. The level of similarity is process. In particular, we focus on the point that the selection of "which constraint to release" directs the reorganization of cognition.

- **As interpretation**: Impasse corresponds to Nami (a state in which prediction-reality discrepancy accumulates and cannot be processed within existing frameworks). Constraint relaxation corresponds to En (discovery of a new relationship — the selection of "which constraint to remove" enables new connections). Representational change and the Aha experience correspond to Uzu (spontaneous establishment of a new coherence). The initial representation is Ba, and consolidation of the updated representation is Taba. This correspondence with the five stages is the most structurally vivid among the 10 theories.
- Furthermore, this theory has a distinctive contribution. While other theories describe En in terms of "what was encountered," insight research describes En in terms of "what was released (constraint relaxation)." This theory most clearly demonstrates that En has not only the aspect of "binding" but also the aspect of "releasing." Also notable is that, using eye movements as an objective measure, it is possible to identify the timing of "when the relationship was formed" — a rare achievement in empirical research on cognitive processes.

Key Entry 2: Bayesian Brain Hypothesis / Predictive Processing (Clark; Friston)

- The Bayesian brain hypothesis is the theory that the brain constantly generates predictions from a world model and continuously updates the model in the direction of minimizing discrepancy (prediction error) with sensory input. Tracing back to Helmholtz's (1867) "unconscious inference," it was formulated as predictive coding by Rao & Ballard (1999) and presented by Clark (2013, 2016) as an integrative framework for cognitive science as a whole.
- **As fact**: Clark (2013) argued that the brain maintains a hierarchical generative model and repeatedly compares predictions with input at each level. Friston (2010) pointed out two pathways for processing prediction errors: one that updates the model itself (perceptual inference), and one that changes the environment through action (active inference). Furthermore, the weighting of prediction errors (precision weighting) is said to be the computational implementation of attention (Feldman & Friston, 2010).
- **As reading**: Here we read the entire process in which, from the generation of prediction error through weighting, the model is updated and stabilized. The level of similarity is structure. In particular, we focus on precision weighting — the selection mechanism of "which error to weight" — as something that not merely processes errors but directs the way relationships are formed.
- **As interpretation**: The generative model (internal representation of the world) corresponds to Ba; the generation of prediction errors corresponds to Nami; precision weighting (the selection of which error to attend to) corresponds to En; the stabilization of a new predictive structure through model update or active inference corresponds to Uzu; and the consolidation of the updated world model corresponds to Taba. This theory demonstrates the possibility of re-describing each phase of the five stages in the vocabulary of computational theory.

Key Entry 3: Schema Theory (Bartlett; Piaget; Neisser)

- Schema theory is the theory that knowledge organizational structures (schemas) extracted from past experience guide the interpretation of new experiences. Bartlett (1932) introduced the schema concept into memory research, Piaget (1952) formulated two processing modes — assimilation (incorporating new information into existing schemas) and accommodation (transforming the schemas themselves) — and Neisser (1976) described a perceptual cycle in which schemas direct exploration and information from the environment modifies schemas.
- **As fact**: In Piaget's framework, when new information is encountered, a distinction is made between processing that incorporates it directly into existing knowledge structures (assimilation) and processing that reworks the knowledge structure itself (accommodation). Which processing is selected depends on the degree of discrepancy between existing schemas and new information.
- **As reading**: Here we read the process in which two processing pathways branch in response to discrepancy between existing knowledge structures and new information, and schemas are

reconstructed through accommodation. The level of similarity is structure. In particular, we focus on the branching point of "which way to process — assimilation or accommodation" as determining the way relationships are formed.

- **As interpretation**: The existing schema (world model) corresponds to Ba; the discrepancy (disequilibrium) between schema and new information corresponds to Nami; the selection branching between assimilation/accommodation (the decision of which way to process) corresponds to En; schema reconstruction through accommodation corresponds to Uzu; and consolidation of the updated schema corresponds to Taba. This theory describes En as "the branching of the way relationships are formed," which corresponds at the cognitive level to the prediction error processing of the Bayesian brain hypothesis.

Cross-Cutting Patterns

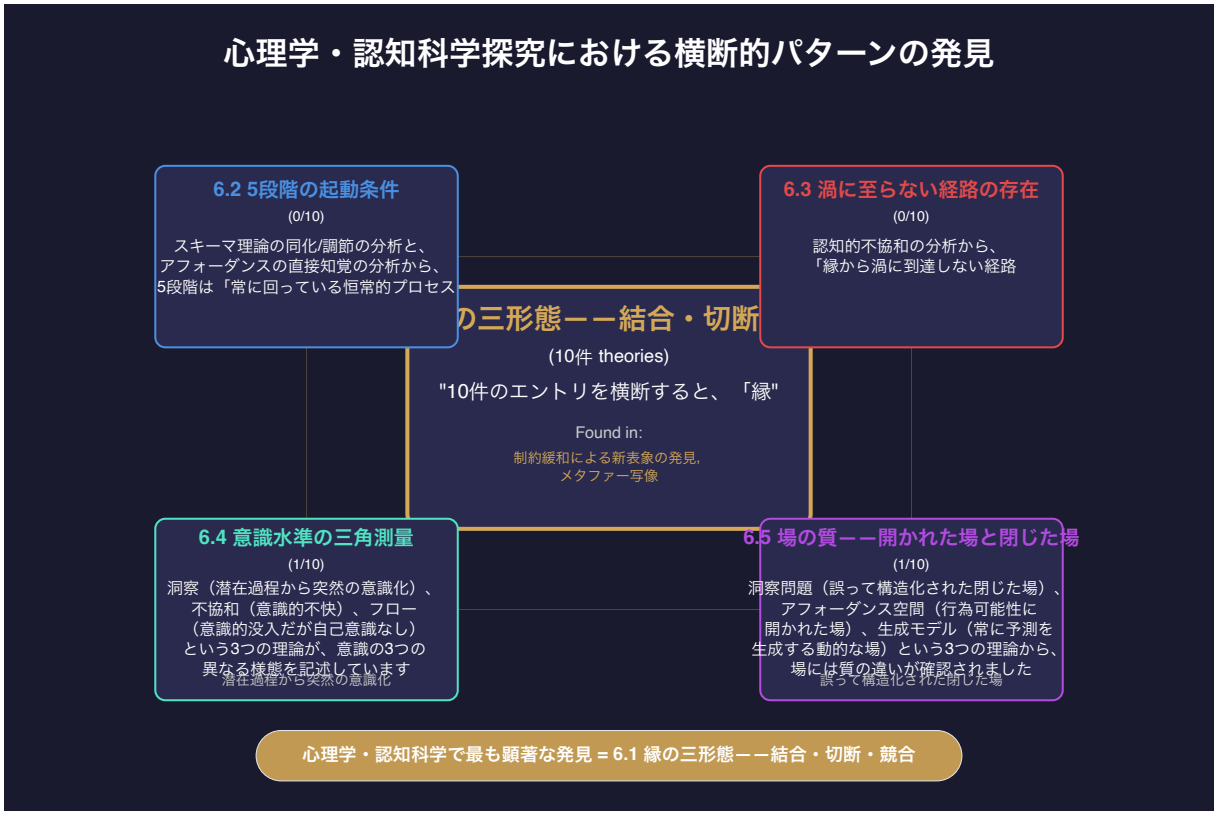

Figure 3: Domain — Cross-Cutting Patterns Diagram

- The most striking pattern across psychology is the **diverse implementations of En**
- **Constraint release**: Insight research describes En from the aspect of release — "which constraint to remove"
- **Distribution of weight**: The Bayesian brain hypothesis describes En through selection — "which error to weight"
- **Branching of processing pathway**: Schema theory describes En through the branching of processing modes — "assimilation or accommodation"

Unresolved Questions

- Whether the diverse implementations of En (constraint release, precision weighting, assimilation/accommodation branching, competition resolution, set switching, strategy selection) share common structural features, or whether the label "En" is being overgeneralized, remains without conclusion.

- Gibson's direct perception theory (no representations) and the conflict with insight and schema theory (representational change as core) relates to the fundamental question of whether "representation" is necessary for the mechanism of En. This opposition has not been resolved within this domain.
- The self-referential question raised by conceptual metaphor theory — if the vocabulary of the five-stage model itself depends on image schemas, does the five-stage model describe the structure of the world, or the structure of understanding via bodily metaphor? — is held in reserve as a question exceeding the scope of this survey.
- The point that En in flow experience (the feedback loop) is relatively automatic and low in indeterminacy relates to the question of how much indeterminacy is required as a definitional condition of En.

Conclusion

- This survey confirmed that psychology and cognitive science is a domain with strong structural similarity to the five-stage model overall
- The five theories showing near-definitive correspondence were: insight problem solving, the Bayesian brain hypothesis, schema theory, the bias competition model of attention, and cognitive flexibility